Scaling the Kitaev/VQE Simulation to Large L

Exact diagonalization, density matrices, matrix product states, and free fermions

Seminar notes

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Abstract

Can the Block-4 noise study be pushed from $L = 4$ toward $L = 16$ (and beyond) on HPC? This note works through the question honestly. The pipeline hides *two* exponential objects with very different escape routes. The **exact-diagonalization reference** is fully removable: the Kitaev Hamiltonian is a free-fermion model, so its ground state and observables follow from a $2L \times 2L$ Bogoliubov–de Gennes (BdG) matrix in $O(L^3)$ time, exact to $L \sim$ hundreds. The **noisy VQE-prepared state** is the hard one: the `EfficientSU2` ansatz is a non-Gaussian, essentially universal circuit, so no classical method escapes exponential cost unconditionally. Matrix product states (MPS) help *when entanglement is low*, and a direct probe of the real ansatz shows the bond dimension χ is bounded *independent of* L at fixed depth ($\chi = 8$ for reps $= 3$, all L). But the same probe exposes a fundamental wall: $\chi \approx 2^{\text{reps}}$ exactly, while the fixed-depth ansatz *loses fidelity* with L ($0.99 \to 0.02$ over $L = 4 \to 12$). Keeping the VQE faithful forces reps $\sim L$, which sends χ back to $2^{O(L)}$. MPS is cheap precisely where the ansatz fails and expensive precisely where it works — there is no free lunch, and that *is* the NISQ statement.

1 Two exponential objects

The current Block-4 evaluation (`src/block4.py`, plots 8–9) contains two objects that scale exponentially in the number of qubits L:

1. the **ED reference** — `np.linalg.eigh` on the $2^L \times 2^L$ Hamiltonian, used for the ideal $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$ curve, the parity gap, and the fidelity post-selection;
2. the **noisy state** — an exact density matrix ρ of dimension $2^L \times 2^L$ (i.e. 4^L complex entries), produced by `save_density_matrix` on the Aer `density_matrix` method.

The observables themselves are *not* the problem: `qubit_hamiltonian`, `edge_string`, and `parity` are already `SparsePauliOp` objects with $O(L)$ Pauli terms. The blow-up comes entirely from calling `.to_matrix()` on them and from `eigh`. The two objects are addressed separately below.

2 The dense wall

Object 2 sets the practical ceiling. A dense $2^L \times 2^L$ complex128 matrix costs $16 \cdot 4^L$ bytes, and the pipeline holds several copies (ρ, H, $\hat{P}$, ED eigenvectors, Aer's internal state):

L	density matrix	+ copies	ED `eigh` cost
12	0.27 GB	∼1 GB	seconds
14	4.3 GB	∼20 GB	minutes
15	17 GB	∼70 GB	∼10 min
16	69 GB	∼300 GB	hours

Worse, *every* VQE cost evaluation is a full 4^L density-matrix simulation, run maxiter × n_starts times per L. Beyond $L \approx 14$ a single L-point becomes hours to days. Even on a large-memory HPC node the dense path does not reach $L = 16$ in feasible wall-clock time; this is the "dense wall" at $L \approx 15$.

3 Escaping object 1: free fermions

The Kitaev Hamiltonian is quadratic in fermions. After a Jordan–Wigner map the qubit form is

$$H = -\tfrac{\mu}{2}\sum_j Z_j + \tfrac{t-\Delta}{2}\sum_j X_j X_{j+1} + \tfrac{t+\Delta}{2}\sum_j Y_j Y_{j+1},$$

which is a free-fermion (BdG) model. Its ground state, spectrum, correlation matrix, and the parity gap $\Delta_P(L) \sim e^{-L/\xi}$ all follow from diagonalizing a $2L \times 2L$ BdG matrix in $O(L^3)$ time — *no* exponential object. The edge operator $X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ is a product of Majorana operators, so its ground-state expectation is a Pfaffian of a submatrix of the covariance matrix (Wick's theorem). This is exactly how Majorana splitting is computed for $L \sim 1000$ in the literature. **Conclusion: the ED reference is removable outright**, exact and cheap to $L \sim$ hundreds. This is the content of `src/free_fermion.py`.

4 Escaping object 2: trajectories and MPS

4.1 Quantum-trajectory statevector sampling

Noise (a mixed state) does not require the full density matrix. In the quantum-trajectory picture each shot is a *pure* statevector (2^L, not 4^L) in which the Kraus operators are sampled stochastically; averaging an observable over N_{traj} trajectories reconstructs $\mathrm{Tr}(O\rho)$ up to a statistical error $\propto 1/\sqrt{N_{\mathrm{traj}}}$. Memory drops from 4^L to 2^L (∼1 MB at $L = 16$), at the honest cost of *error bars* on the noisy curves.

4.2 Matrix product states

A general L-qubit state is a tensor with 2^L amplitudes. An MPS factorizes it into a chain of small tensors, one per site, linked by bond indices of dimension χ:

$$|\psi\rangle = \sum_{\{s\}} A^{s_0} A^{s_1} \cdots A^{s_{L-1}} |s_0 \dots s_{L-1}\rangle, \qquad \text{memory} = O(L\,\chi^2).$$

Across any cut, χ equals the number of nonzero Schmidt coefficients — a direct measure of *entanglement* across that cut. A gapped 1D ground state obeys an area law, so its entanglement is bounded independent of L; then χ is constant and MPS is *linear* in L. A critical or volume-law state needs $\chi \sim 2^{L/2}$ and MPS is exponential again. Aer provides `method='matrix_product_state'` and supports noise through trajectory sampling; the empirical test confirms it runs and exposes the per-bond χ.

4.3 Truncation error, and why the risk is controllable

A two-qubit gate can double χ across its bond. To stay cheap one performs an SVD there and keeps only the largest $\chi_{\max}$ Schmidt values. The discarded weight — the sum of squared discarded Schmidt coefficients — *is* the truncation error: the fraction of the state's norm thrown away. If nothing is discarded the MPS is *exact*. If $\chi_{\max}$ is set below what the state needs, the simulated state becomes a nearby but different state and $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$ acquires a systematic bias. Crucially this is a knob with a readout: Aer reports the discarded weight, and the convergence check "run at $\chi_{\max}$ and $2\chi_{\max}$; if the observable does not move, the result is effectively exact" makes the error visible and quantitative rather than hidden. A truncated number never silently corrupts a result as long as the discarded weight is logged.

5 The χ-probe: what the real ansatz actually does

We measured the *exact* (uncapped) max bond dimension of the real `EfficientSU2`(`ry`, linear, reps) state directly.

χ is flat in L at fixed depth. Worst-of-random-θ max χ (reps = 3):

L	6	10	14	18	22
χ	8	8	8	8	8
$2^{\lfloor L/2 \rfloor}$ (vol. law)	8	32	128	512	2048

A reps $= r$ linear ansatz has a light cone of width $\sim r$, so entanglement across any cut is bounded *independent of* L. At reps $= 3$, χ pins at 8 for all L, far below the volume-law ceiling, and noise does not inflate it. **The truncation risk essentially does not exist for a shallow ansatz**: cap $\chi_{\max} = 16$ and the simulation is exact to $L = 100$.

χ grows exponentially in depth. At fixed $L = 14$:

reps	1	2	3	4	5	6	8
χ	2	4	8	16	32	64	128

i.e. $\chi \approx 2^{\text{reps}}$ (until it saturates the $2^{L/2}$ ceiling).

The fixed-depth ansatz loses the state. VQE fidelity to the true ground state at reps = 3:

L	4	6	8	10	12
fidelity	0.99	0.50	0.25	0.10	0.02

A fixed-depth hardware-efficient ansatz simply cannot represent the topological ground state as L grows; keeping the fidelity up requires reps $\sim L$.

6 The wall is fundamental, not a coding artifact

Combining the three tables: the regime where MPS is cheap (small reps, constant χ) is exactly the regime where the VQE *fails* (fidelity $\to 0$); and the regime where the VQE *works* (reps $\sim L$) is exactly where $\chi \sim 2^{\text{reps}} \sim 2^{O(L)}$ blows MPS back up. A state that genuinely needs a deep circuit to prepare is genuinely expensive to simulate classically — MPS buys a smaller exponential base (2^{reps} vs 4^L), a few extra qubits, not unbounded L. This is the honest simulation-cost face of "NISQ does not inherit topological protection."

The one unconditionally polynomial regime. The true Kitaev ground state is Gaussian and area-law: it *has* a small-χ MPS. The `ry` ansatz is merely an inefficient, non-Gaussian parametrization of it. A **matchgate/Gaussian ansatz** (only R_z-type single-qubit gates + nearest-neighbor matchgate entanglers) prepares the same state at $O(L)$ depth with bounded χ, and under *fermion-Gaussian* noise (amplitude damping/loss is Gaussian for fermions; depolarizing is not) the entire simulation reduces to covariance-matrix algebra, $O(L^3)$, exact, $L \to$ thousands. The catch: this answers a cleaner but *different* question — it is no longer the generic hardware-efficient NISQ story.

7 Recommended architecture

piece	scalable replacement	exact?
ideal $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$, gap	free-fermion BdG (L^3)	exact, $L \to$ hundreds
noisy VQE $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$	MPS + noise trajectories ($L\chi^2$)	exact if χ bounded
unbounded L	matchgate ansatz + Gaussian noise (L^3)	exact, different model

The concrete plan: build `src/free_fermion.py` (an unconditional win) and pair it with a *fixed-reps* MPS sweep that makes the ansatz's expressibility collapse visible against the exact ideal curve. That is honest, cheap, reaches large L, and turns the wall itself into the scientific point. Validation anchor: at $L \leq 12$, where the exact density-matrix path still runs, cross-check the MPS-trajectory and free-fermion values against the exact numbers before extrapolating.

Backends for the noisy curve. A device-calibrated sweep needs a snapshot whose coupling map embeds a length-L line so the linear-entanglement `cx` pairs sit on real edges with no SWAP inflation. Named 5-qubit snapshots (`fake_manila`, etc.) cap at $L = 5$; `FakeGuadalupeV2` is 16 qubits but its heavy-hex graph only contains a 13-node line. The clean choices are the 27-qubit heavy-hex snapshots (`FakeCairoV2`/`FakeKolkataV2`/`FakeHanoiV2`, 21-node line) or a 20-qubit device (18-node line).

Reproduce. χ-probe: `python scratch/chi_probe.py` (measures χ vs L and vs reps against the volume-law ceiling). Free-fermion reference: `src/free_fermion.py` (BdG ground state, edge string, parity gap; validated against `eigh` at $L \leq 12$).